Advanced Machine Learning

Lecture 10: Residual Neural Networks & Variants

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Outline

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Long-term dependencies

Simple RNN suffer from one major problem: that of long-term dependencies.

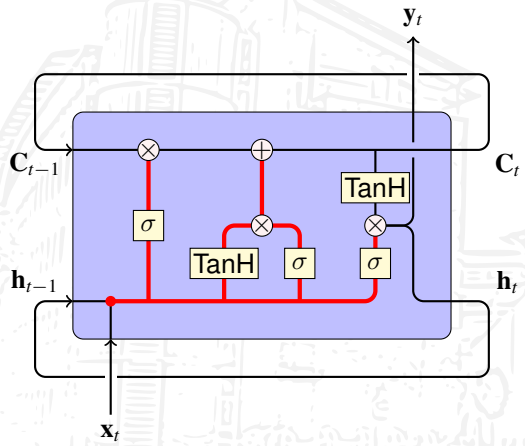
As the hidden state is transformed in every step by a matrix multiplication, long-term dependencies are hard to learn. The classical example is a sentence like

> *I was born in France. [. . .] I speak French.*

The longer the part in brackets, the smaller the probality that the RNN learns this long-term dependency. RNN can only learn dependencies that are as long ago as the unrolling parameter permits. In many cases RNN do not learn dependencies that are too long-term; simple RNN only have a short-term memory.

The solution to deal with long-term dependencies are Long Short-Term Memory (LSTM) networks. These have been published in Hochreiter and Schmidhuber [5].

LSTM



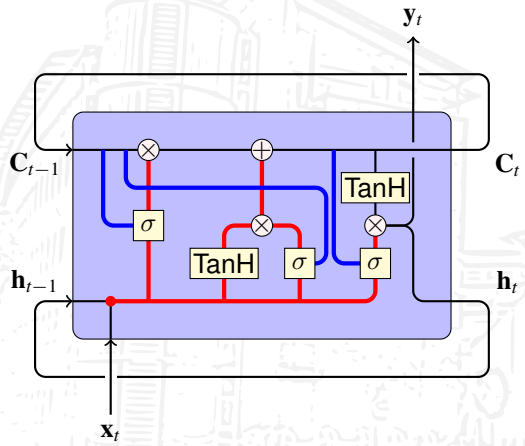
LSTM have a hidden state $\mathbf{h}_t$ and an additional cell state $\mathbf{C}_t$ that stores long-term dependencies.

The inner structure is more complicated than for a simple RNN, we have

- five classical layers w/ weight matrices, biases, thrice the logistic function σ, two activation functions, here twice TanH,
- three componentwise multiplications and one componentwise addition,
- one vector concatenation, five copies.

In this example the output $\mathbf{y}_t$ is equal to the hidden state $\mathbf{h}_t$.

LSTM variants: peephole connections



There exist variants that introduce peepholes that allow the gates to see the cell state, e.g., the forget gate looks as follows:

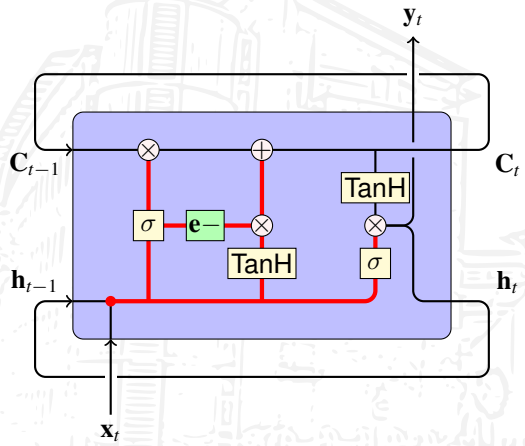
$$\mathbf{f}_t = \sigma\left(\mathbf{W}_f \begin{pmatrix} \mathbf{C}_{t-1} \\ \mathbf{h}_{t-1} \\ \mathbf{x}_t \end{pmatrix} + \mathbf{b}_f\right),$$

where $\mathbf{W}_f$ splits into

$$\mathbf{W}_f = \begin{pmatrix} \mathbf{W}_{fC} & \mathbf{W}_{fh} & \mathbf{W}_{fx} \end{pmatrix}.$$

This is a generalization: In case $\mathbf{W}_{fC} = \mathbf{O}$ we recover the variant w/o peephole.

LSTM variants: coupled input and forget gates



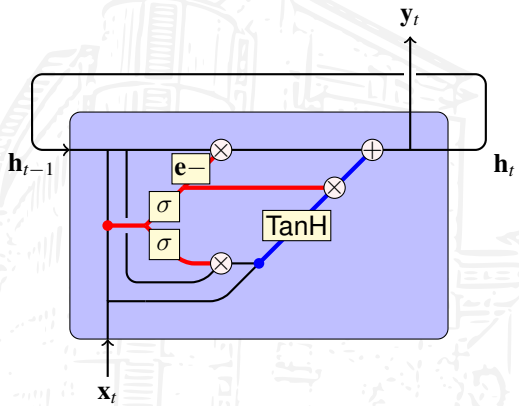
Using a model with limited memory, we insert something new when we forget something. Thus, a variation on LSTM is to couple the input and forget gates,

$$\mathbf{C}_t = \mathbf{C}_{t-1} \circ \mathbf{f}_t + \widetilde{\mathbf{C}}_t \circ (\mathbf{e} - \mathbf{f}_t),$$

where $\mathbf{e}$ is the vector of all ones.

Here, $\mathbf{i}_t = \mathbf{e} - \mathbf{f}_t$ and we compute the new cell state as a convex combination of the old cell state and the new candidate cell state.

LSTM variants: GRU



The Gated Recurrent Unit (GRU) by Cho et al. [1] is a more dramatic variation of an LSTM.

Here, cell and hidden state are merged, input and forget gate are coupled, and no bias is used,

$$\mathbf{z}_t = \sigma\left(\mathbf{W}_z \begin{pmatrix} \mathbf{h}_{t-1} \\ \mathbf{x}_t \end{pmatrix}\right), \quad \mathbf{r}_t = \sigma\left(\mathbf{W}_r \begin{pmatrix} \mathbf{h}_{t-1} \\ \mathbf{x}_t \end{pmatrix}\right),$$

$$\widetilde{\mathbf{h}}_t = \text{TanH}\left(\mathbf{W}_h \begin{pmatrix} \mathbf{h}_{t-1} \circ \mathbf{r}_t \\ \mathbf{x}_t \end{pmatrix}\right),$$

$$\mathbf{h}_t = (\mathbf{e} - \mathbf{z}_t) \circ \mathbf{h}_{t-1} + \mathbf{z}_t \circ \widetilde{\mathbf{h}}_t.$$

Only two gates: update gate $\mathbf{z}_t$ & reset gate $\mathbf{r}_t$.

BPTT for LSTM

We consider the defining equations of an LSTM and derive the corresponding backpropagation through time.

The LSTM is based on the three gates (forget gate, input gate, output gate),

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_f \begin{pmatrix}\mathbf{h}_{t-1}\\ \mathbf{x}_t\end{pmatrix} + \mathbf{b}_f\right), \quad \mathbf{i}_t = \sigma\left(\mathbf{W}_i \begin{pmatrix}\mathbf{h}_{t-1}\\ \mathbf{x}_t\end{pmatrix} + \mathbf{b}_i\right), \quad \mathbf{m}_t = \sigma\left(\mathbf{W}_h \begin{pmatrix}\mathbf{h}_{t-1}\\ \mathbf{x}_t\end{pmatrix} + \mathbf{b}_h\right),$$

the cell state update

$$\mathbf{C}_t = \mathbf{C}_{t-1} \circ \mathbf{f}_t + \widetilde{\mathbf{C}}_t \circ \mathbf{i}_t, \quad \widetilde{\mathbf{C}}_t = \mathrm{TanH}\left(\mathbf{W}_C \begin{pmatrix}\mathbf{h}_{t-1}\\ \mathbf{x}_t\end{pmatrix} + \mathbf{b}_C\right),$$

and the hidden state update that depends on the previous hidden state,

$$\mathbf{h}_t = \mathrm{TanH}(\mathbf{C}_t) \circ \mathbf{m}_t.$$

BPTT for LSTM

Like in the simple RNN case we consider a cost function, here $\mathbf{y}_t = \mathbf{h}_t$,

$$C(\{\mathbf{x}_t\}_{t=0}^k, \{\mathbf{y}_t\}_{t=0}^k) := \sum_{t=0}^{k} C_t(\{\mathbf{x}_\ell\}_{\ell=0}^t, \mathbf{y}_t).$$

The hidden state evolves as follows,

$$\mathbf{h}_t = \mathrm{TanH}(\mathbf{C}_t) \circ \mathbf{m}_t = f(\mathbf{h}_{t-1}).$$

For the last hidden state in step $t = k$ we can easily compute

$$\frac{\partial C}{\partial \mathbf{h}_k} = \frac{\partial C_k}{\partial \mathbf{h}_k},$$

for the previous steps $t < k$ we end up with the recursion

$$\frac{\partial C}{\partial \mathbf{h}_t} = \frac{\partial C_t}{\partial \mathbf{h}_t} + \frac{\partial \mathbf{h}_{t+1}}{\partial \mathbf{h}_t} \cdot \frac{\partial C}{\partial \mathbf{h}_{t+1}}.$$

BPTT for LSTM

A similar recursion develops for the cell state $\mathbf{C}_t$, which occurs in the two formulas

$$\mathbf{C}_{t+1} = \mathbf{C}_t \circ \mathbf{f}_{t+1} + \widetilde{\mathbf{C}}_{t+1} \circ \mathbf{i}_{t+1}, \quad \mathbf{h}_t = \mathsf{TanH}(\mathbf{C}_t) \circ \mathbf{m}_t.$$

In the last step $t = k$ we only have to compute

$$\frac{\partial C}{\partial \mathbf{C}_k} = \frac{\partial \mathbf{h}_k}{\partial \mathbf{C}_k} \cdot \frac{\partial C}{\partial \mathbf{h}_k} = \Big(\mathbf{e} - \mathsf{TanH}^2(\mathbf{C}_k)\Big) \circ \mathbf{m}_k \circ \frac{\partial C}{\partial \mathbf{h}_k},$$

in previous steps $t < k$ we have the recurrence

$$\frac{\partial C}{\partial \mathbf{C}_t} = \frac{\partial \mathbf{h}_t}{\partial \mathbf{C}_t} \cdot \frac{\partial C}{\partial \mathbf{h}_t} + \frac{\partial \mathbf{C}_{t+1}}{\partial \mathbf{C}_t} \frac{\partial C}{\partial \mathbf{C}_{t+1}} = \Big(\mathbf{e} - \mathsf{TanH}^2(\mathbf{C}_t)\Big) \circ \mathbf{m}_t \circ \frac{\partial C}{\partial \mathbf{h}_t} + \mathbf{f}_{t+1} \circ \frac{\partial C}{\partial \mathbf{C}_{t+1}}.$$

Thus, in an LSTM, we have two recurrences, one for the hidden state, one for the cell state.

BPTT for LSTM

We define

$$\mathbf{a}_t := \begin{pmatrix} \mathbf{f}_t \\ \mathbf{i}_t \\ \mathbf{m}_t \\ \widetilde{\mathbf{C}}_t \end{pmatrix}, \quad \mathbf{z}_t := \begin{pmatrix} \mathbf{z}_{ft} \\ \mathbf{z}_{it} \\ \mathbf{z}_{ht} \\ \mathbf{z}_{Ct} \end{pmatrix} := \begin{pmatrix} \mathbf{W}_{fh}\,\mathbf{h}_{t-1} + \mathbf{W}_{fx}\,\mathbf{x}_t + \mathbf{b}_f \\ \mathbf{W}_{ih}\,\mathbf{h}_{t-1} + \mathbf{W}_{ix}\,\mathbf{x}_t + \mathbf{b}_i \\ \mathbf{W}_{hh}\,\mathbf{h}_{t-1} + \mathbf{W}_{hx}\,\mathbf{x}_t + \mathbf{b}_h \\ \mathbf{W}_{Ch}\mathbf{h}_{t-1} + \mathbf{W}_{Cx}\mathbf{x}_t + \mathbf{b}_C \end{pmatrix} =: \mathbf{W}_h\mathbf{h}_{t-1} + \mathbf{W}_x\mathbf{x}_t + \mathbf{b}.$$

We have (thrice logistic function, once hyperbolic tangent)

$$\frac{\partial \mathbf{a}_t}{\partial \mathbf{z}_t} = \operatorname{diag} \begin{pmatrix} (\mathbf{e} - \mathbf{f}_t) \circ \mathbf{f}_t \\ (\mathbf{e} - \mathbf{i}_t) \circ \mathbf{i}_t \\ (\mathbf{e} - \mathbf{m}_t) \circ \mathbf{m}_t \\ \mathbf{e} - \widetilde{\mathbf{C}}_t \circ \widetilde{\mathbf{C}}_t \end{pmatrix}, \quad \frac{\partial \mathbf{z}_t}{\partial \mathbf{h}_{t-1}} = \mathbf{W}_h^\mathsf{T}, \quad \frac{\partial \mathbf{z}_t}{\partial \mathbf{x}_t} = \mathbf{W}_x^\mathsf{T}.$$

On the next slide we consider the three gates and the candidate state by looking at the partial derivatives with respect to $\mathbf{z}_{ft}$, $\mathbf{z}_{it}$, $\mathbf{z}_{ht}$, and $\mathbf{z}_{Ct}$.

BPTT for LSTM

We use

$$\mathbf{C}_t = \mathbf{C}_{t-1} \circ \mathbf{f}_t + \widetilde{\mathbf{C}}_t \circ \mathbf{i}_t, \quad \mathbf{h}_t = \mathsf{TanH}(\mathbf{C}_t) \circ \mathbf{m}_t$$

to compute the derivatives with respect to the gates and candidate cell state,

$$\begin{aligned}
\frac{\partial C}{\partial \mathbf{z}_{ft}} &= \frac{\partial \mathbf{f}_t}{\partial \mathbf{z}_{ft}} \frac{\partial \mathbf{C}_t}{\partial \mathbf{f}_t} \frac{\partial C}{\partial \mathbf{C}_t} = (\mathbf{e} - \mathbf{f}_t) \circ \mathbf{f}_t \circ \mathbf{C}_{t-1} \circ \frac{\partial C}{\partial \mathbf{C}_t}, \\
\frac{\partial C}{\partial \mathbf{z}_{it}} &= \frac{\partial \mathbf{i}_t}{\partial \mathbf{z}_{it}} \frac{\partial \mathbf{C}_t}{\partial \mathbf{i}_t} \frac{\partial C}{\partial \mathbf{C}_t} = (\mathbf{e} - \mathbf{i}_t) \circ \mathbf{i}_t \circ \widetilde{\mathbf{C}}_t \circ \frac{\partial C}{\partial \mathbf{C}_t}, \\
\frac{\partial C}{\partial \mathbf{z}_{ht}} &= \frac{\partial \mathbf{m}_t}{\partial \mathbf{z}_{ht}} \frac{\partial \mathbf{h}_t}{\partial \mathbf{m}_t} \frac{\partial C}{\partial \mathbf{h}_t} = (\mathbf{e} - \mathbf{m}_t) \circ \mathbf{m}_t \circ \mathsf{TanH}(\mathbf{C}_t) \circ \frac{\partial C}{\partial \mathbf{h}_t}, \\
\frac{\partial C}{\partial \mathbf{z}_{Ct}} &= \frac{\partial \widetilde{\mathbf{C}}_t}{\partial \mathbf{z}_{Ct}} \frac{\partial \mathbf{C}_t}{\partial \widetilde{\mathbf{C}}_t} \frac{\partial C}{\partial \mathbf{C}_t} = (\mathbf{e} - \widetilde{\mathbf{C}}_t \circ \widetilde{\mathbf{C}}_t) \circ \mathbf{i}_t \circ \frac{\partial C}{\partial \mathbf{C}_t}.
\end{aligned}$$

BPTT for LSTM

Now we can compute the missing part that will be handed to the previous time step,

$$\frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_{t-1}} \frac{\partial C}{\partial \mathbf{h}_t} = \frac{\partial \mathbf{z}_t}{\partial \mathbf{h}_{t-1}} \frac{\partial C}{\partial \mathbf{z}_t} = \mathbf{W}_h^\mathsf{T} \frac{\partial C}{\partial \mathbf{z}_t},$$

and the change in the input (handed to the previous layer)

$$\frac{\partial C}{\partial \mathbf{x}_t} = \frac{\partial \mathbf{z}_t}{\partial \mathbf{x}_t} \frac{\partial C}{\partial \mathbf{z}_t} = \mathbf{W}_x^\mathsf{T} \frac{\partial C}{\partial \mathbf{z}_t}.$$

The update of weights and biases (shared over time) is given by

$$\frac{\partial C}{\partial \mathbf{W}_h} = \sum_{t=0}^{k} \frac{\partial C}{\partial \mathbf{z}_t} \mathbf{h}_{t-1}^\mathsf{T}, \quad \frac{\partial C}{\partial \mathbf{W}_x} = \sum_{t=0}^{k} \frac{\partial C}{\partial \mathbf{z}_t} \mathbf{x}_t^\mathsf{T}, \quad \frac{\partial C}{\partial \mathbf{b}} = \sum_{t=0}^{k} \frac{\partial C}{\partial \mathbf{z}_t}.$$

BPTT for LSTM

The implementation of BPTT for LSTM is based on transferring the state given by

$$\mathbf{W}_h^\mathsf{T} \frac{\partial C}{\partial \mathbf{z}_t}, \quad \mathbf{f}_t \circ \frac{\partial C}{\partial \mathbf{C}_t}$$

to the previous step and computing all partial derivatives for the current time step, updating the derivatives of the weights and biases.

As optimizers all variants of SGD can be used, e.g., Adam.

A list of examples that show results obtained using LSTM can be found at Andrej Karpathy's Blog:

http://karpathy.github.io/2015/05/21/rnn-effectiveness/

Outline

Networks we already know

We already know the following types of neural networks:

Fully Connected Neural Networks: dense layers are activated by various activation functions

Convolutional Neural Networks: convolutional layers replace some/all of the dense layers; weights are shared over spatial coordinates

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Unfortunately, this does not work out of the box.

Can we go deeper?

When we use one of the net structures we have and add layers, the training takes longer, the learning rate has to be decreased, and the final accuracy will not be as good as for less layers.

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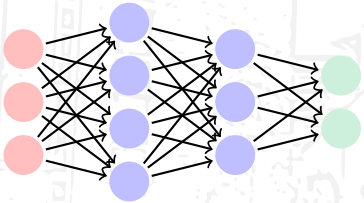

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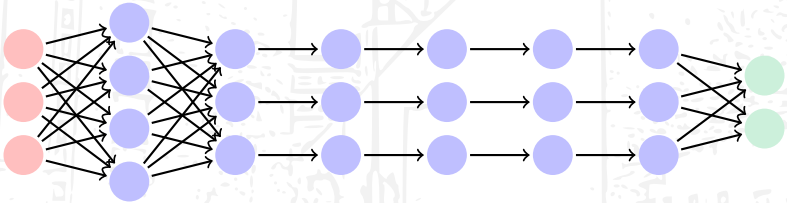

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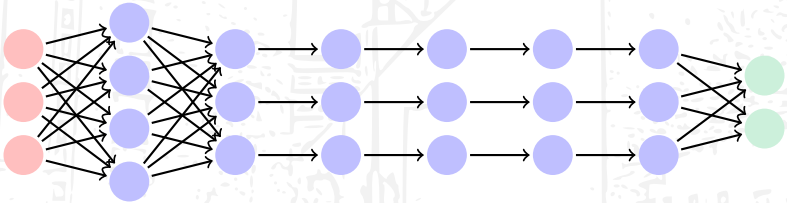

Thus, a trained net of that form should at least be as good as the smaller one.

The problem

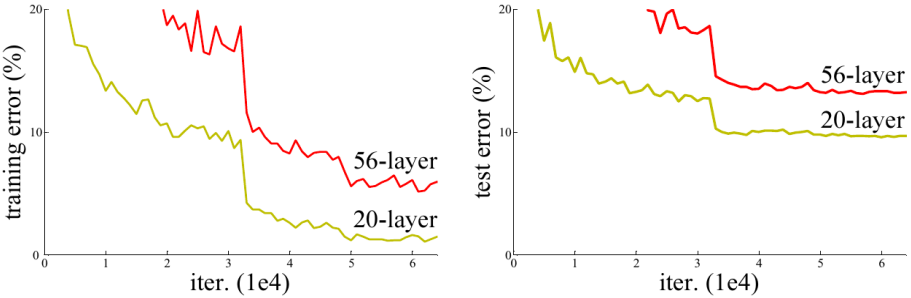

Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer "plain" networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

Image taken from He et al. [3].

Where the problem occurs

Amount of layers in networks:

A few hidden layers: suitable choosen initialization, learning rate, minibatch size mostly suffice

Tens of layers: additionally batch normalization, weight decay, or other regularization has to be used to address the problem of vanishing/exploding gradients

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Nets with hundreds of layers trainable by backpropagation using SGD exist. How do they look like?

Ideas to cope with the problem

In LSTM gates enable to keep information over long periods of steps. The idea of using gates can be adapted to feedforward and convolutional networks, see Srivastava, Greff, and Schmidhuber [8] (the extended version is Srivastava, Greff, and Schmidhuber [9]):

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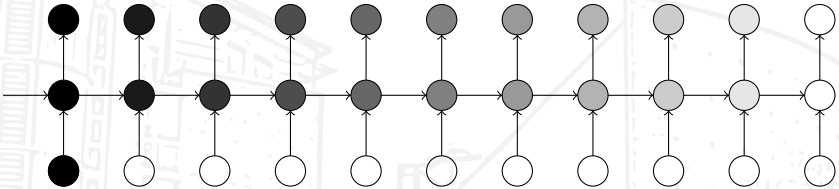
> *"Yes, there are two paths you can go by, but in the long run, there's still time to change the road you're on."*

(Led Zeppelin, Stairway to Heaven, 8. November 1971)

Gates in feedforward/convolutional nets

In the PhD thesis of Graves [2] LSTM are explained as follows. A simple RNN gradually looses information:



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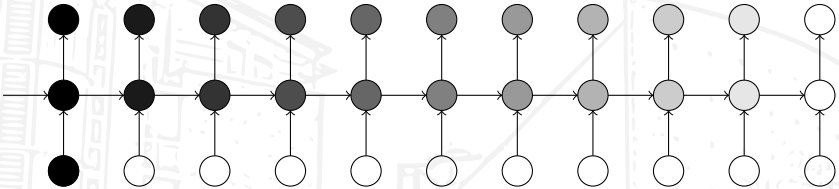
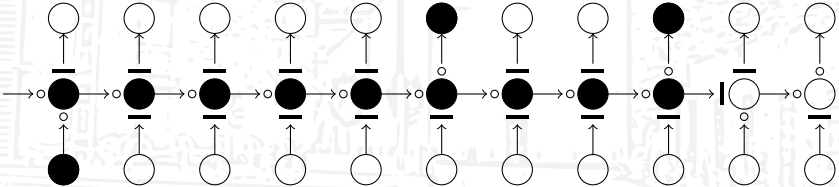


An LSTM can keep the information longer due to the three gates:



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In Highway Networks we replace a layer H that maps input to output of equal dimension, e.g.,

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$$\mathbf{y} = H(\mathbf{x}) = a(\mathbf{W}\mathbf{x} + \mathbf{b}), \qquad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n,$$

by including a transform gate T and a carry gate C (often these are coupled, $C = \mathbf{e} - T$),

$$T(\mathbf{x}) = \sigma(\mathbf{W}_t\mathbf{x} + \mathbf{b}_t), \quad C(\mathbf{x}) = \sigma(\mathbf{W}_c\mathbf{x} + \mathbf{b}_c),$$

to obtain, similar to the cell state update in an LSTM, the mapping

$$\mathbf{y} = H(\mathbf{x}) \circ T(\mathbf{x}) + \mathbf{x} \circ C(\mathbf{x}).$$

Results for Highway Networks

They performed experiments that showed “that highway networks as deep as 900 layers can be optimized using simple SGD with momentum.”

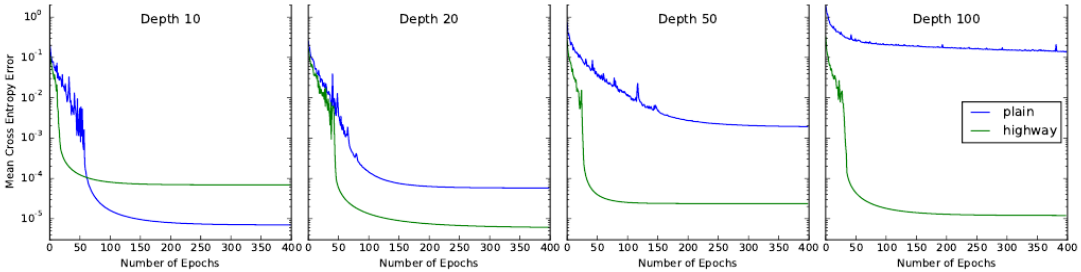

Figure 1. Comparison of optimization of plain networks and highway networks of various depths. All networks were optimized using SGD with momentum. The curves shown are for the best hyperparameter settings obtained for each configuration using a random search. Plain networks become much harder to optimize with increasing depth, while highway networks with up to 100 layers can still be optimized well.

Image taken from Srivastava, Greff, and Schmidhuber [8].

Results for Highway Networks ($C = \mathbf{e} - T$)

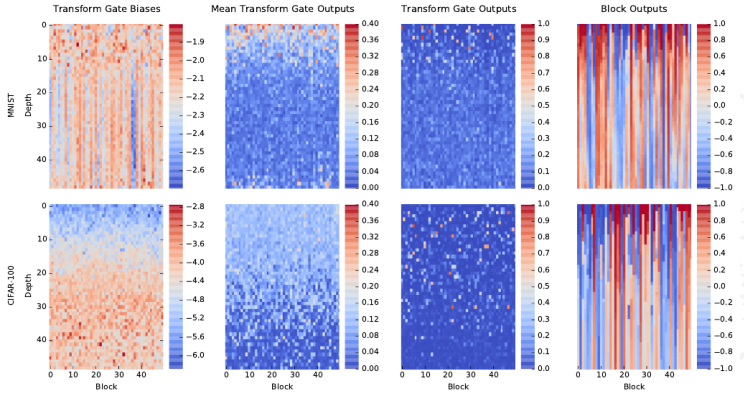

Figure 2. Visualization of certain internals of the blocks in the best 50 hidden layer highway networks trained on MNIST (top row) and CIFAR-100 (bottom row). The first hidden layer is a plain layer which changes the dimensionality of the representation to 50. Each of the 49 highway layers (y-axis) consists of 50 blocks (x-axis). The first column shows the transform gate biases, which were initialized to -2 and -4 respectively. In the second column the mean output of the transform gate over 10,000 training examples is depicted. The third and forth columns show the output of the transform gates and the block outputs for a single random training sample.

Image taken from Srivastava, Greff, and Schmidhuber [8].

Results for Highway Networks ($C = \mathbf{e} - T$)

Initially, the bias of the transformation gate is set to a negative number such that the gate is closed and the carry gate is open, here -2 for the MNIST set and -4 for the CIFAR-100 set. This ensures that in the beginning the data flows through the net mostly unchanged.

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In MNIST, lesioning the layers 15–45 seems to have no effect (MNIST is very simple dataset). In CIFAR-100 the effect is more dramatic in earlier layers; the effect decays linearly in the layer number.

Results for Highway Networks ($C = \mathbf{e} - T$)

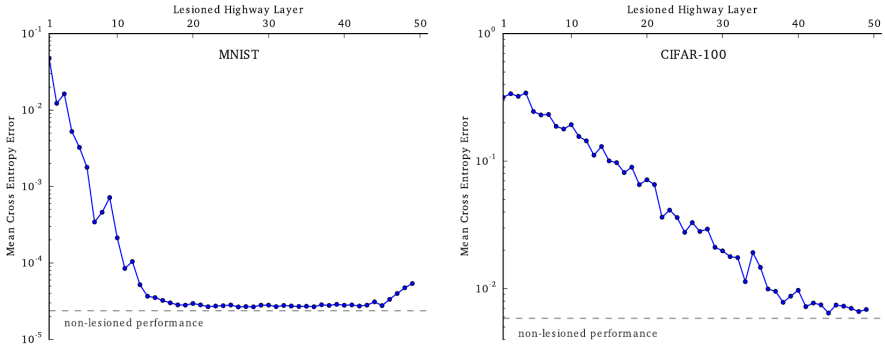

Figure 4: Lesioned training set performance (y-axis) of the best 50-layer highway networks on MNIST (left) and CIFAR-100 (right), as a function of the lesioned layer (x-axis). Evaluated on the full training set while forcefully closing all the transform gates of a single layer at a time. The non-lesioned performance is indicated as a dashed line at the bottom.

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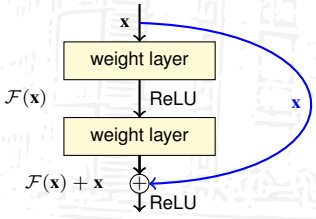
The formulation of $\mathcal{F}(\mathbf{x}) + \mathbf{x}$ is realized by a feedforward neural network with shortcut connections:



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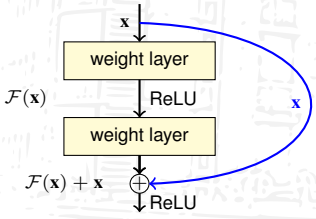


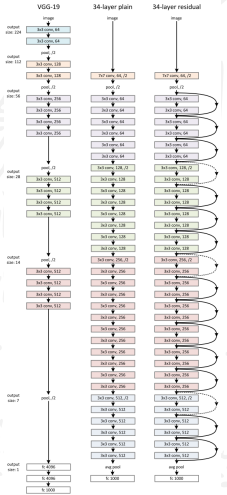
In this example, which is used in He et al. [3],

$$\mathcal{F}(\mathbf{x}) := \mathbf{W}_2 \mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$$

is followed by the addition and then the activation,

$$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x}, \quad \mathbf{z} = \mathrm{ReLU}(\mathbf{y}).$$

Definition of ResNet-34



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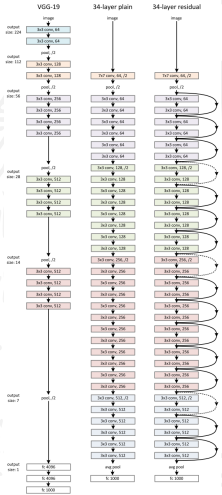

The difference between the plain-34 net and the ResNet-34 in He et al. [3] lies in the additional 16 shortcut connections. For comparison they included in the first column the VGG-19 net with 19 layers.

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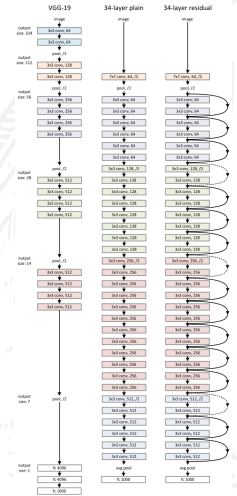

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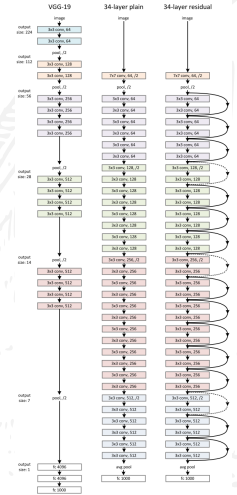

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Three of the shortcut connections are based on padding the input by zeros, as the dimension increases, these are the ones with the dashed arrows in the picture. Such transformations have already been defined in Srivastava, Greff, and Schmidhuber [9].

Results for ResNet-{18,34}

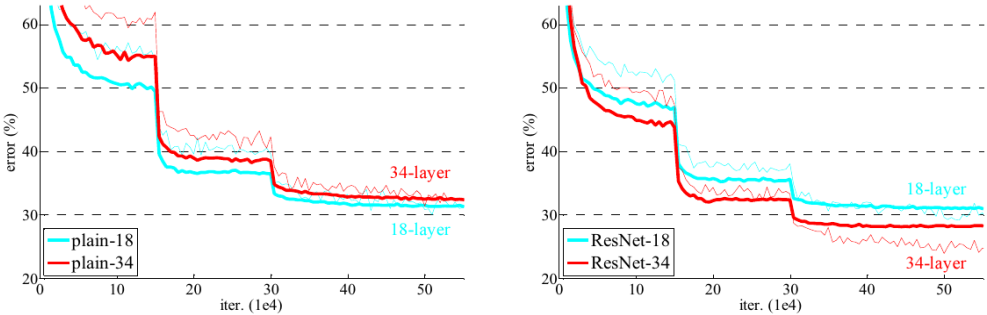

Figure 4. Training on **ImageNet**. Thin curves denote training error, and bold curves denote validation error of the center crops. Left: plain networks of 18 and 34 layers. Right: ResNets of 18 and 34 layers. In this plot, the residual networks have no extra parameter compared to their plain counterparts.

Image taken from He et al. [3].

Variants of ResNet (scramble BN, ReLU, affine linear)

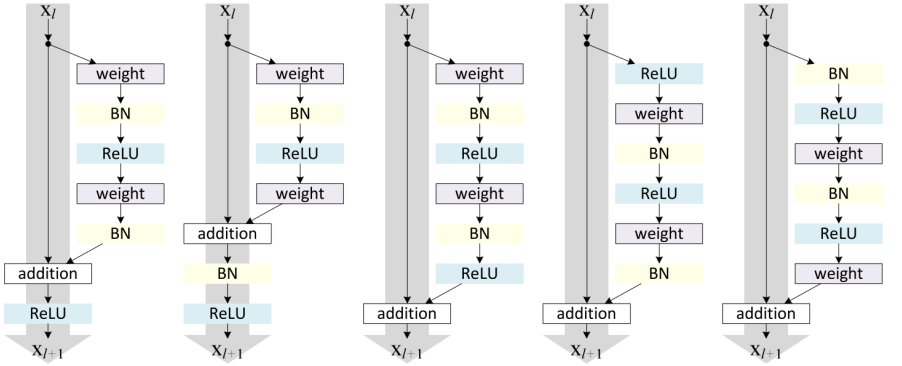

Image taken from He et al. [4].

ResNet-1001, ResNeXt

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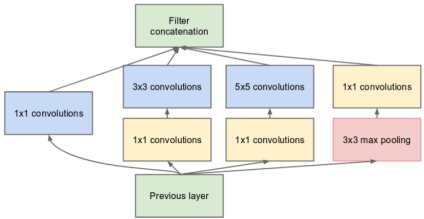
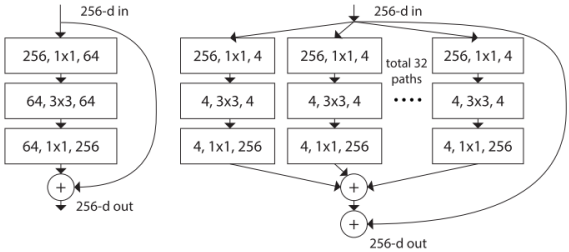
ResNeXt: Xie et al. [12] combine ResNet with the Inception modules of GoogLeNet V1 from Szegedy et al. [10],



(b) Inception module with dimension reductions

Image taken from Szegedy et al. [10].



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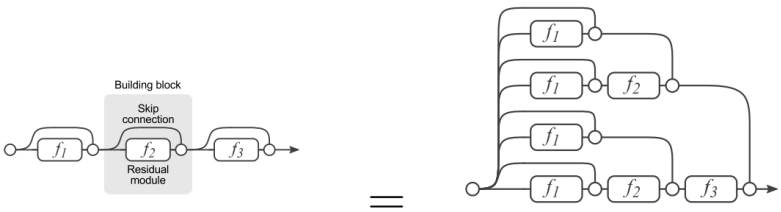

(a) Conventional 3-block residual network

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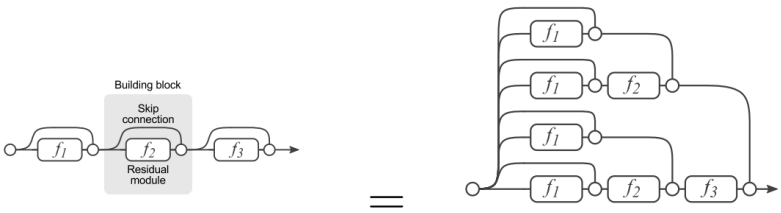

(a) Conventional 3-block residual network (b) Unraveled view of (a)

Image taken from Veit, Wilber, and Belongie [11].

They investigate how removal of a residual block changes the performance. In the next plot the arising lesioning is shown.

How many paths do we need?

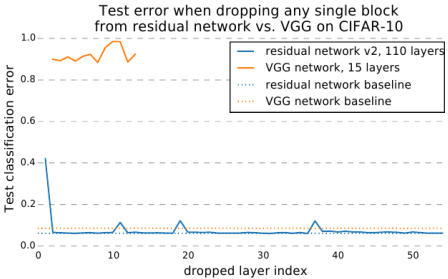

Figure 3: Deleting individual layers from VGG and a residual network on CIFAR-10. VGG performance drops to random chance when any one of its layers is deleted, but deleting individual modules from residual networks has a minimal impact on performance. Removing downsampling modules has a slightly higher impact.

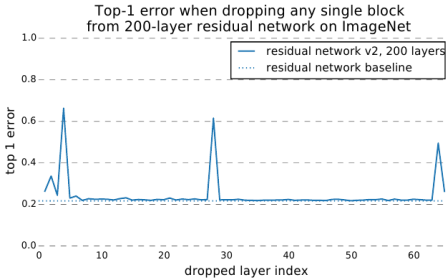

Figure 4: Results when dropping individual blocks from residual networks trained on ImageNet are similar to CIFAR results. However, downsampling layers tend to have more impact on ImageNet.

Image taken from Veit, Wilber, and Belongie [11].

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At test time, all layers are used with the residual block multiplied by p_ℓ,

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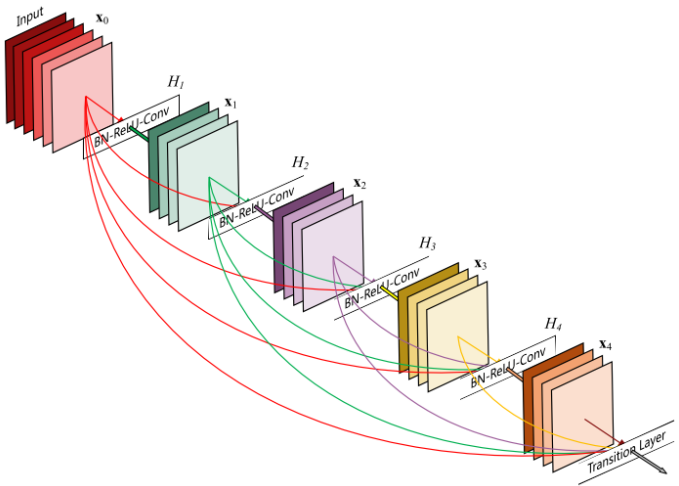
They successfully trained a ResNet with 1202 layers based on training w/ stochastic depth.

Adding more paths



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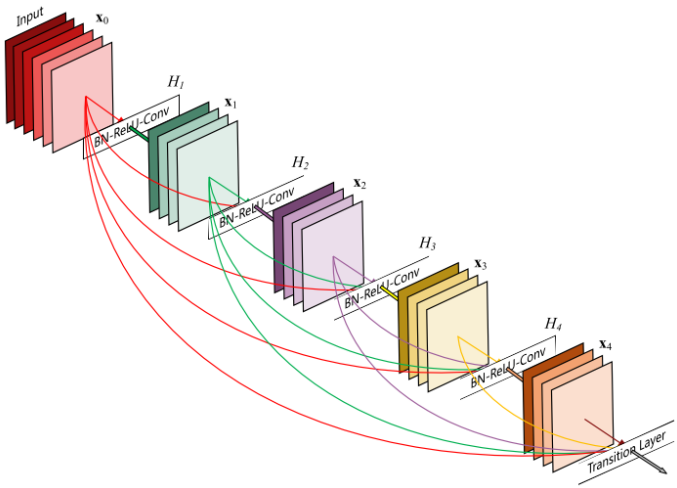

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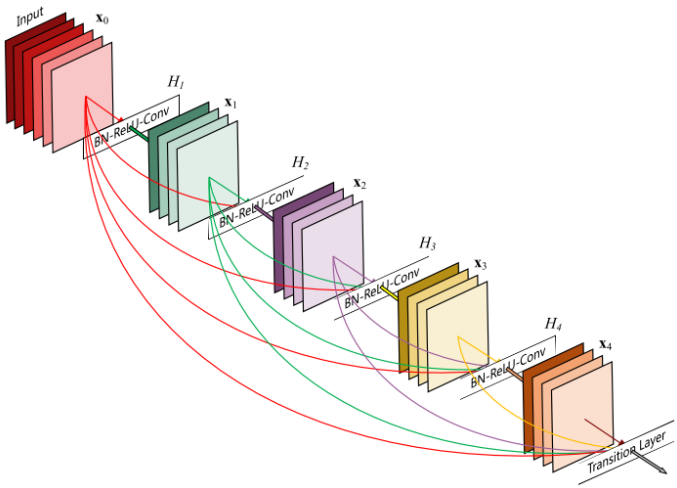

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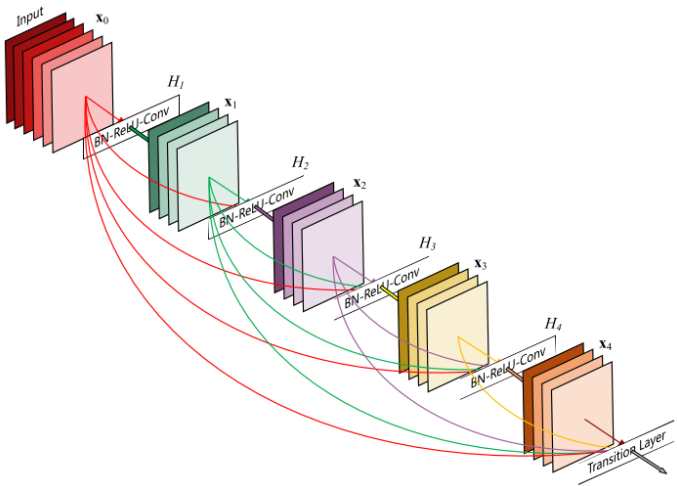

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In contrast to a ResNet, the inputs are concatenated.

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